Arjuna JEE 2027

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Quadratic Equations

DPP: 6

Total Duration - 51 Min

Q1 The quadratic equations $x^2-6x+a=0$ and $x^2-cx+6=0$, have one root in common. The other roots of the first and second equations are integers in the ratio $4:3$. Then, the common root is

(A) 2 (B) 1

(C) 4 (D) 3

Q2 If α and β are the roots of the equation, $7x^2-3x-2=0$, then the value of $\frac{\alpha}{1-\alpha^2}+\frac{\beta}{1-\beta^2}$ is equal to

[JEE Main 2020, 5 Sep (Shift 2)]

(A) $\frac{3}{8}$

(B) $\frac{1}{24}$

(C) $\frac{27}{16}$

(D) $\frac{27}{32}$

Q3 Let p, q and r be real numbers $(p\neq q, r\neq 0)$, such that the roots of the equation $\frac{1}{x+p}+\frac{1}{x+q}=\frac{1}{r}$ are equal in magnitude but opposite in sign, then the sum of squares of these roots is equal to:

(A) $\frac{p^2+q^2}{2}$

(B) p^2+q^2

(C) $2\left(p^2+q^2\right)$

(D) $p^2+q^2+r^2$

Q4 If the roots of the equation $x^2-8x+\left(a^2-6a\right)=0$ are real, then

(A) $-2<a<8$

(B) $2<a<8$

(C) $-2\leq a\leq 8$

(D) $2\leq a\leq 8$

Q5 The least positive value of 'a' for which the equation, $2x^2+(a-10)x+\frac{33}{2}=2a$ has real roots is ______

Q6 If the roots of the quadratic equation $x^2-ax+2b=0$ are prime numbers, then the value of $(a-b)$ is

(A) 0 (B) 2

(C) -2 (D) 4

Q7 The roots of the equation $(3-x)^4+(2-x)^4=(5-2x)^4$ are

(A) All real

(B) All imaginary

(C) Two real and two imaginary

(D) None of these

Q8 If α, β are the roots of quadratic equation $x^2+px+q=0$ and γ, δ are the roots of $x^2+px-r=0$, then $(\alpha-\gamma).(\alpha-\delta)$ is equal to :

(A) $q+r$ (B) $q-r$

(C) $-(q+r)$ (D) $-(p+q+r)$

Q9 If α, β are roots of the equation $ax^2+bx+c=0$ then the equation whose roots are $2\alpha+3\beta$ and $3\alpha+2\beta$ is

(A) $abx^2-(a+b)cx+(a+b)^2=0$

(B) $acx^2-(a+c)bx+(a+c)^2=0$

(C) $acx^2(a+c)bx-(a+c)^2=0$

(D) None of these

Q10 Let α, β be the roots of the equation $x^2+ax+b=0$ and γ, δ be the roots of $x^2-ax+b-2=0$. If $\alpha\beta\gamma\delta=24$ and $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}+\frac{1}{\delta}=\frac{5}{6}$, then find the value of a.

Q11 If the quadratic equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0 (a \neq 0)$ have a common root, then the numerical value of $(a + b)$ is

(A) 0 (B) 1

(C) -1 (D) 2

Q12 If the two equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$ have one common root and the second has equal roots, then $2(b + d)$ is equal to

(A) 0 (B) $a + c$

(C) ac (D) $-ac$

Q13 If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0, a, b, c \in R$, have a common root, then $a : b : c$ is:

(A) $1 : 2 : 3$ (B) $3 : 2 : 1$

(C) $1 : 3 : 2$ (D) $3 : 1 : 2$

Q14 $x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, if $a =$

(A) 24 (B) $0, 24$

(C) $3, 24$ (D) $0, 3$

Q15 If $(x + k)$ is a common factor of the expressions $x^2 + px + q$ and $x^2 + lx + m$, then k is equal to

(A) $\frac{p+q}{\ell+m}$

(B) $\frac{p-\ell}{q-m}$

(C) $\frac{q-m}{p+\ell}$

(D) $\frac{q-m}{p-\ell}$

Q16 If the equation $x^2 + ax + b = 0$ and $x^2 + a'x + b' = 0$ have a common root, then this common root is equal to

(A) $\frac{(b-b')}{(a-a')}$

(B) $\frac{(b+b')}{(a'-a)}$

(C) $\frac{(b-b')}{(a+a')}$

(D) $\frac{(b-b')}{(a'-a)}$

Q17 The number of roots of the equation $\sqrt{x-3}\left(x^2 - 7x + 10\right) = 0$ is

(A) 2 (B) 3

(C) Zero (D) 1

Q18 If $1, 2, 3$ are the roots of the equation $x^3 + ax^2 + bx + c = 0$, then

(A) $a = 1, b = 2, c = 3$

(B) $a = -6, b = 11, c = -6$

(C) $a = 6, b = 11, c = 6$

(D) $a = 6, b = 6, c = 6$

Q19 The roots of the equation $x^3 - 2x^2 - x + 2 = 0$ are

(A) $1, 2, 3$ (B) $-1, 1, 2$

(C) $-1, 0, 1$ (D) $-1, -2, 3$

Q20 If $x = 3 + 3^{2/3} + 3^{1/3}$, then the value of the expression $x^3 - 9x^2 + 18x - 10$ is equal to ________.

Answer Key

Q1	(A)
Q2	(C)
Q3	(B)
Q4	(C)
Q5	8
Q6	(B)
Q7	(C)
Q8	(C)
Q9	(D)
Q10	10
Q11	(C)
Q12	(C)
Q13	(A)
Q14	(B)
Q15	(D)
Q16	(D)
Q17	(A)
Q18	(B)
Q19	(B)
Q20	2